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Self-healing, highly elastic and amphiphilic silicone-based polyurethane for antifouling coatings†

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Silicone elastomer coatings have attracted increasing attention owing to their eco-friendly nature, excellent fouling release ability and drag-reducing property. However, the poor mechanical properties and lack of fouling resistance limit their applications. Herein, a silicone-based polyurethane with 2-ureido-4[1H]-pyrimidinone (UPy) and amphiphilic pendant chains is reported. The UPy groups and urethane can form hydrogen bonds with various substrates so that the coating has significantly improved adhesion strength (0.9–3.0 MPa) compared with the PDMS elastomer (0.3–0.4 MPa). Moreover, the quadruple hydrogen bonding between UPy moieties allows the polymer to have excellent self-healing ability and high elasticity. The modified polymeric coating has low surface energy (24 mJ m⁻²) and low elastic modulus (1.9 MPa), so it exhibits good fouling release performance. Besides, the amphiphilic side chains can effectively resist protein adsorption and adhesion of the marine bacteria *Pseudomonas* sp. and diatom *Navicula incerta*. The novel design can improve the resistance of silicone coating to biofouling and mechanical forces and is promising for high-performance antifouling coatings.

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Introduction

Marine biofouling is the undesired settlement and accumulation of marine organisms on materials submerged in seawater.¹ It is detrimental to the marine industry, increasing drag force and fuel consumption, causing the loss of vessel maneuverability, and accelerating corrosion and species invasion.² Many antifouling technologies have been developed to combat biofouling, including fouling release coatings (FRCs),^{3,4} degradable polymers with dynamic surfaces,^{5–7} biomimetic materials,⁸ protein-resistant polymers,⁹ amphiphilic polymers^{10–12} and so on.¹³ Of these, coatings with toxic biocides are the most widely used antifouling strategy. However, the released biocides represent tremendous hazard to the marine ecological environment.¹⁴

FRCs have received wide attention for their environmental and drag-reducing advantages.¹⁵ Polydimethylsiloxane (PDMS) elastomers are widely used in FRCs because of their low surface energy (SE) and modulus that make fouling organisms easily detach with hydrodynamic stress or mechanical cleaning.¹⁶ However, they easily foul under static conditions and have poor substrate adhesion as the polysiloxane chains are non-polar. Besides, they are easily damaged by an external force, and their

cross-linked networks cannot be repaired, all of which limit their antifouling applications. To improve antifouling performances under static conditions, fouling-resistant moieties like amphiphiles,¹⁷ quaternary ammonium salts,¹⁸ and zwitterions¹⁹ have been incorporated into silicone. Introducing these groups can lead to a decline in the adhesion and mechanical strength of the polymer because most of these moieties are hydrophilic, which accelerates the swelling of silicones. To improve these properties, PDMS can be physically modified with nanofillers like carbon nanotubes,²⁰ and graphene oxide,²¹ or chemically modified with epoxy²² or polyurethane.^{23,24} However, these modifications lead to an increase in the SE and elastic modulus, reducing the fouling release ability. Although these modified silicones have improved antifouling or adhesion properties, their irreparable nature remains unresolved. To date, self-healing materials have been widely researched using built-in capsules or reversible interactions such as hydrogen bonds,²⁵ Diels–Alder bonds,²⁶ disulfide bonds,^{27,28} and others.²⁹ Particularly, quadruple hydrogen bonding formed by 2-ureido-4[1H]-pyrimidinone (UPy) has attracted increasing attention since it is promising in developing high performance self-healing materials, such as highly elastic self-healing hydrogels, supramolecular polymer materials for thin film electrodes and so on.^{30–32} So far, only a few studies have focused on self-healing silicones for marine anti-biofouling as many self-healing polymers have no inherent fouling-resistant abilities. Antifoulants or silicone oils are still

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required to improve the antifouling performances, perpetuating adverse effects on the marine ecosystem.³³ It is imperative to design a non-leaching, self-healing system with improved adhesion and fouling resistance.

In this study, we have prepared a silicone-based polyurethane with UPy and amphiphilic pendant chains. The presence of urethane bonds is expected to increase adhesion to the substrate, and the reversible quadruple hydrogen bonds of UPy further improve adhesion and add a self-healing capacity. On the other hand, the chemically attached amphiphiles can enrich the surface coating because of the incompatibility of the fluorocarbon segments with silicone and improve antifouling performances. We examined the adhesion, mechanical properties, self-healing performance and wettability of the coatings as well as studied their resistance to proteins, marine bacteria, and diatoms, and their fouling release ability, with the goal of developing environmentally friendly, high-performance antifouling coatings.

Experimental section

Materials

Difunctional carbinol-terminated poly(dimethylsiloxane) (PDMS, $M_w = 1800 \text{ g mol}^{-1}$) was purchased from Shin-Etsu Chemical Co., Ltd, and dried under reduced pressure at 110°C for 2 h prior to use. 2-Amino-6-methyl-4-pyrimidinol was purchased from Alfa Aesar. 2-Amino-2-methyl-1,3-propanediol (AMPD) from Sigma Aldrich was used as received. Hexamethylene diisocyanate (HDI) was purchased from TCI and used as received. 1H,1H,7H-Dodecafluoroheptyl methacrylate (DFMA) was purchased from Harbin Xuejia Fluorin Silicon Chemical Co., Ltd. Poly(ethylene glycol) methyl ether methacrylate (PEGMA, $M_n = 300 \text{ g mol}^{-1}$) from Sigma Aldrich and 3-mercapto-1,2-propanediol (TPG) from TCI were used as received. Isophorone diisocyanate (IPDI), 1,4-butanediol (BDO), dibutyltin dilaurate (DBTDL), dibutyltin diacetate (DBTDA) and ethyl acetate (EA) from Aladdin were used as received. 2,2'-Azobis(2-methylpropionitrile) (AIBN) from Aladdin was recrystallized twice from methanol. Bis-silanol terminated PDMS ($M_n = 26000 \text{ g mol}^{-1}$) and ethyl silicate 40 (ES40) were purchased from Gelest and used as received. Fibrinogen (fraction I from human plasma, $M_w = 340 \text{ kDa}$, pI 5.5) was purchased from Merck Chemicals. Tetrahydrofuran (THF) and hexane from Sinopharm were purified by distillation after stirring with CaH_2 . Anhydrous dimethylformamide (DMF) was purchased from Energy Chemical. Artificial seawater (ASW) was prepared following ASTM D1141-98 (2003).

Synthesis of amphiphilic telomer

The amphiphilic telomer was synthesized *via* free radical polymerization (Fig. S1, ESI[†]). As reported before,^{11,12,17} PEG segments are readily buried in the bulk of low surface energy PDMS coatings. Therefore, fluorocarbon segments are needed to drag the PEG segments onto the surface, forming a hydrophilic and fouling-resistant surface. Typically, DFMA (2.40 g, 6 mmol), PEGMMA (3.60 g, 12 mmol), TPG (0.32 g, 3 mmol),

and AIBN (0.03 g, 0.5 wt%) were dissolved in THF (21.0 mL) and added to a Schlenk tube with a magnetic stir bar. The solution was filled with N_2 after three freeze-pump-thaw degassing cycles. The solution was then allowed to react at 70°C for 24 h while stirring and the resulting solution was precipitated with hexane. A viscous solution was obtained after removing the residual solvent at 60°C under reduced pressure. ^1H NMR (600 MHz, CDCl_3 , ppm, Fig. S2, ESI[†]): 0.85–1.02 (CH_3CCH_2), 1.84–1.96 (CH_3CCH_2), 4.10 (OCH_2CH_2), 3.60 ($\text{CH}_2\text{CH}_2\text{O}$), 2.52 (OCH_3), 4.60 (OCH_2CF_2), 5.50 (CF_2CHF_2), 2.58 (CHOHCH_2S), 4.21 (CHOHCH_2S), 4.16 (HOCH_2CH). IR: 3450 (OH), 2950 (CH_3), 1730 (C=O). M_n and polydispersity index (PDI) of the telomer determined by gel permeation chromatography (GPC) are 1800 g mol^{-1} and 1.82, respectively.

Synthesis of UPy-diol

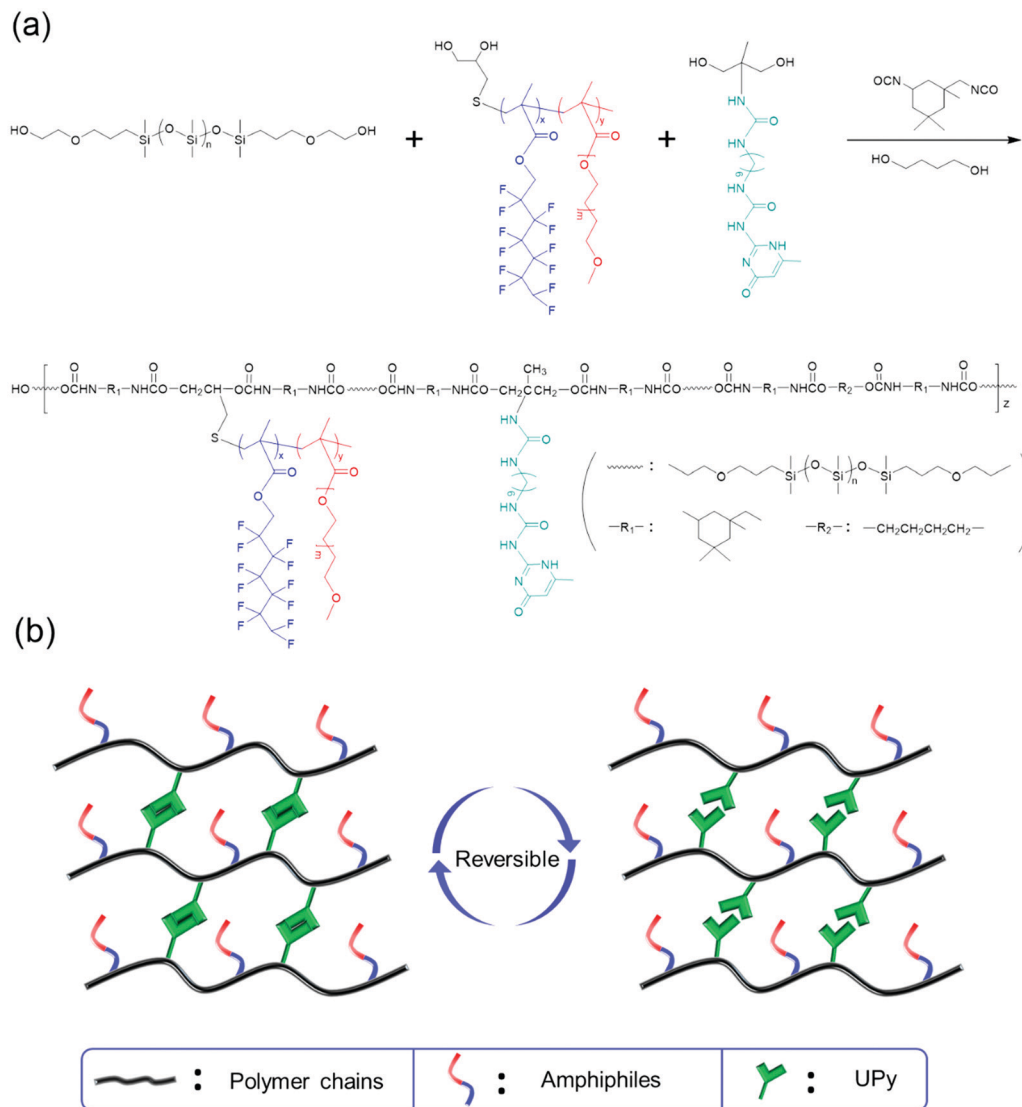
The synthesis of UPy-diol was performed by two steps according to the methods in the literature (Fig. S3, ESI[†]).³⁴

(1) **Synthesis of 2-(6-isocyanato-hexylamino)-6-methyl-4[1H]-pyrimidinone (UPy-NCO).** Typically, 6.26 g (50 mmol) of 2-amino-6-methyl-4-pyrimidinol and 50.46 g (300 mmol) of HDI were added to a 250 mL three-neck round bottom flask with a magnetic stir bar. The mixture was reacted at 100°C for 24 h under a N_2 environment. After cooling to room temperature, the mixture was washed by hexane three times to remove unreacted HDI. The product was dried in a vacuum oven at 50°C to obtain a white powder. ^1H NMR (600 MHz, CDCl_3 , ppm, Fig. S4, ESI[†]): 13.11 (CH_3CNH), 11.86 (CNHCONH), 10.18 (CONHCH_2), 5.82 (CH_3CCH), 3.29 ($\text{NHCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NCO}$), 2.23 (CH_3CNH), 1.62 ($\text{NHCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NCO}$), 1.40 ($\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NCO}$).

(2) **Synthesis of UPy-diol.** After the synthesis of UPy-NCO, 13.96 g (47.7 mmol) of UPy-NCO was mixed with 7.36 g (70.0 mmol) of AMPD in CH_3Cl . The mixture was stirred at 60°C for 5 h under a N_2 atmosphere. The product was then filtered and washed with CH_3Cl , followed by dissolution in anhydrous DMF to obtain a cloudy solution. The cloudy solution was centrifuged to isolate the supernatant liquid. After that, the supernatant liquid was precipitated by excess diethyl ether, and a white product was obtained by filtration. The white product was washed with acetone several times. ^1H NMR (600 MHz, $\text{DMSO}-d_6$, ppm, Fig. S5, ESI[†]): 9.87 (CH_3CNH), 8.24 (CNHCONH), 6.17 (NHCONHCCH_3), 5.71 (NHCONHCCH_3), 5.64 (CH_3CCH), 5.11 ($\text{OHCH}_2\text{CCH}_3$), 3.47–3.26 ($\text{NHCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}$), 3.13 ($\text{OHCH}_2\text{CCH}_3$), 2.93 ($\text{OHCH}_2\text{CCH}_3$), 2.10 (CH_3CNH), 1.2–1.5 ($\text{NHCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}$), 1.08 ($\text{OHCH}_2\text{CCH}_3$).

Preparation of silicone-based polyurethane

Silicone-based polyurethanes with different telomer contents were synthesized *via* addition reaction (Scheme 1). As we know, polyurethane can be successfully synthesized by the prepolymer method or one-pot method.^{35,36} Generally, a more uniform polyurethane structure could be obtained using the prepolymer method.²⁸ Therefore, silicon-based polyurethane was synthesized by the prepolymer method and the reaction degree was



Scheme 1 (a) Synthetic route and (b) schematic illustration of the self-healing PDMS-UPy-Tx polymer.

determined by titration according to GB/T 12099.4-2016. Considering that a small part of NCO could be consumed with moisture in the air, a protocol of excess of NCO was employed.³⁴ The molar ratio of NCO/OH was kept at 1.05 to ensure the chain extension was completed. Typically, 10.80 g (6.0 mmol) of difunctional carbinol-terminated PDMS and 4.45 g (20.0 mmol) of IPDI were introduced to a three-neck round bottom flask equipped with a condenser under a N₂ atmosphere with THF as the solvent. The mixture was stirred at 70 °C for 1 h, yielding a prepolymer with excess of free isocyanate. The titration result showed that the degree of pre-polymerization was ~30%. However, the unreacted IPDI can participate in the subsequent polymer reaction. Then, 0.90 g (0.5 mmol) of telomer was added. To increase the reactivity of isocyanate groups with the hydroxyl groups of the telomer, the temperature was raised to 80 °C. The titration result showed that the degree of pre-polymerization was ~38% at 3 h, and it reached the end point at 9 h. Finally, 0.92 g (2.3 mmol) of

UPy-diyl dissolved in anhydrous DMF was added, followed by the addition of 0.93 g (10.3 mmol) of BDO for chain extension. 0.5 wt% DBTDL (based on total monomer) was also added to increase the reactivity of the isocyanate-terminated prepolymer with hydroxyl groups of chain extenders. The solution was allowed to react at 80 °C overnight to ensure the chain extension was completed. The titration result showed that there was no residual NCO in solution after reacting overnight. The resulting polymer was precipitated with excess hexane, washed three times, and dried under vacuum. Polymers with other telomer contents were synthesized by the same protocol. ¹H-NMR (600 MHz, CDCl₃, ppm, Fig. S6, ESI[†]): 0.06 (OSi(CH₃)₂O), 0.52 (CH₂OCH₂CH₂CH₂Si), 1.61 (CH₂OCH₂CH₂CH₂Si), 3.42 (CH₂OCH₂CH₂CH₂Si), 3.60 (COOCH₂CH₂OCH₂), 3.72 (COOCH₂CH₂OCH₂), 1.06 (CH₂C(CH₃)₂CH₂), 1.25 (CH₂CHNHCOO), 3.79 (CH₂CHNHCOO), 2.91 (CH₃CCH₂NHCOO), 3.37 (CH₂CH₂OCH₃), 4.09 (OCH₂CH₂CH₂CH₂O), 1.68 (OCH₂CH₂CH₂CH₂O). IR(KBr) (Fig. S7, ESI[†]): 3330 cm⁻¹ (NH), 2960 cm⁻¹ (CH₃), 1700 cm⁻¹

Table 1 Characterization data of PDMS-UPy-Tx

Sample	PDMS/IPDI/UPy-diol/BDO/telomer ^a	Telomer ^b (wt%)	M_n^c (g mol ⁻¹)	M_w^c (g mol ⁻¹)	D_M^c	T_g^d (°C)
PDMS-UPy-T0	0.53/2/0.20/1.18/0	0	16 000	22 400	1.4	25.6
PDMS-UPy-T5	0.60/2/0.23/1.03/0.05	4.8	16 000	24 000	1.5	23.2
PDMS-UPy-T10	0.70/2/0.27/0.82/0.12	10.1	17 000	23 800	1.4	17.7
PDMS-UPy-T15	0.85/2/0.32/0.53/0.22	14.7	15 000	21 000	1.4	16.6

^a Feed molar ratio. ^b Determined by ¹H NMR. ^c Determined by GPC. ^d Determined by DSC.

(C=O), 1540 cm⁻¹ (C-N), 1256 cm⁻¹ (C-O), 1090 cm⁻¹ (Si-O-Si), 790 cm⁻¹ (Si-C). For convenience, the synthesized polymer is denoted as PDMS-UPy-Tx, where x represents the telomer weight percent determined by ¹H NMR. In this study, 5 wt% of UPy-diol was employed to maintain an identical density of quadruple hydrogen bonds, and different telomer contents were studied. The characterization data are summarized in Table 1.

The polymer was dissolved in THF and applied to various substrates including fiberglass epoxy resin, Q235 steel, glass, concrete and wood (for adhesion test, thickness: ~300 μm), glass slide (for contact angle measurement, self-healing test, and surface elastic modulus measurement), fiberglass epoxy resin (for fouling release test), silicon wafer (for antibacterial and anti-diatom tests), and AT-cut quartz crystal (spin-cast on the surface; for protein resistance test). Polymer solution was cast on a Teflon dish to prepare the polymer film, which was dried at 25 °C in a fume hood for 3 days and in vacuum for another day to remove the solvent completely, and then it was cut into a rectangular shaped sample with 40 mm length, 15 mm width and 0.8 mm thickness (for tensile test). The polymer film with blue or red color was prepared by mixing the polymer solution (3.0 mL) with blue or red ink (0.1 mL) from color pens (for macroscopic healing test). The room temperature vulcanized PDMS was synthesized by condensation reaction.³⁷ Typically, 10.00 g of bis-silanol terminated PDMS, 0.25 g of ES40 and 2.5 mg of DBTDA were dissolved in 30 mL of EA and stirred at room temperature, followed by casting on a Teflon dish for the tensile test or on various substrates for the adhesion test. The coatings were cured at ambient temperature for 3 days and dried under vacuum before conducting the experiment.

Characterization

Proton nuclear magnetic resonance spectroscopy (¹H NMR). The chemical structure was analyzed by ¹H NMR on a Bruker AV600 NMR spectrometer using CDCl₃ or DMSO-d₆ as the solvent.

Gel permeation chromatography (GPC). The number average molecular weight (M_n) and the polydispersity of the polymer were measured on an Agilent 1260 Infinity system with refractive index (RI) and ultraviolet (UV) detectors. The test was performed at 35 °C with two identical PLgel columns (5 μm, MIXED-C) and THF was chosen as the eluent with a flow rate of 1.0 mL min⁻¹. The calibration curve was determined by pre-tests on polystyrene standards (1.30 × 10³ to 2.21 × 10⁶ g mol⁻¹) with narrow polydispersity.

Fourier Transform Infrared Spectroscopy (FTIR). The spectra were recorded on a Bruker VECTOR-22 FTIR spectrometer using the KBr disk method with scanning ranges from 4000 to 500 cm⁻¹ and at 4 cm⁻¹ resolution.

Differential scanning calorimetry (DSC). The glass transition temperature (T_g) of the polymer was measured using a NETZSCH differential scanning calorimeter (Netzsch DSC 204F1). The sample was heated and kept at 120 °C for 5 min to erase thermal history. Then it was cooled down to -80 °C at a rate of 10 °C min⁻¹ and equilibrated for 5 min, and reheated to 120 °C at a rate of 10 °C min⁻¹. The T_g was determined from the midpoint temperature of the endothermic shift during the second heating curve.

Adhesion test. The adhesion strength of PDMS-UPy-Tx and PDMS on fiberglass epoxy resin, concrete, Q235 steel, wood and glass was measured using a Posi Test AT-A Automatic tester according to ASTM D4541. All substrates were roughed using 60-grit abrasive paper (Shanghai Changjiang Grinding Wheel Factory Co., Ltd), cleaned and dried to remove impurities and increase the surface roughness. The coating was prepared by the solution casting method and dried at room temperature and the thickness of the coating was ~300 μm. A 20 mm diameter aluminum dolly was attached to the coating using epoxy adhesive (Araldite Rapid Epoxy Adhesive, Huntsman). The pull rate was set at 0.2 MPa s⁻¹. Five different regions were measured to obtain an average value. The adhesion strength of PDMS-UPy-Tx after conditioning in ASW was evaluated using the same protocol, in which the coating was pre-immersed in ASW for 24 h.

Tensile test. The mechanical properties and self-healing efficiency of PDMS-UPy-Tx elastomers were evaluated by a tensile test using an Instron 5966 instrument with a 100 N load cell at 25 °C. The sample was rectangular shaped with 15 mm width, 0.8 mm thickness and 20 mm gage length. The rate of elongation was 25 mm min⁻¹ and all the tests were repeated at least three times. For evaluating self-healing efficiency, elastomer films were cut using a razor blade, and then reconnected quickly and allowed to heal at different temperature for 48 h. Self-healing efficiency was defined as the tensile strength ratio between healed and original samples. Toughness was calculated by the integration of stress-strain curves. Cyclic tensile tests were performed by stretching the samples to 200% strain, and then unloading to original position using the same instrument at 25 °C with a constant strain rate of 25 mm min⁻¹. Each sample was successively loaded and unloaded with 10 cycles. Energy dissipation was determined from hysteresis loops of individual cycle.

Self-healing test. The macroscopic healing tests of the polymer film were recorded using a digital camera. The self-healing process of coatings in air or ASW was photographed at room temperature using an optical microscope.

Contact angle measurement. Static contact angles for deionized water or diiodomethane on the coatings were measured using a Theta Auto 113 (KSV NIMA, Biolin) at 25 °C by dropping a droplet of liquid (3 μL) on the sample surface using the sessile method. An average value was obtained by measuring five different regions of each sample. The measurement of the dynamic water contact angle (WCA) was performed according to a previous report.³⁸ The advancing WCA is sensitive to the hydrophobic component of the surface, whereas the receding WCA is sensitive to the hydrophilic component of the surface. The surface free energies were obtained using the Owens-Wendt-Rabel-Kaelble method.

Surface elastic modulus measurement. The elastic modulus of the PDMS-UPy-Tx coatings was measured by atomic force microscopy (AFM) (Park Systems, Korea) using Hertzian contact theory, which correlates the AFM force signal (F) to the indentation depth of the cantilever tip (δ) and the elastic modulus (E) by the following equation:³⁹

$$F = \frac{4}{3} \frac{E}{(1 - \nu^2)} \sqrt{R} \delta^{3/2}$$

where ν is the coating's Poisson's ratio and is 0.5 for all coatings, R is the radius of the spherical tip, and δ is the indentation depth.

Fouling release test. The removal strength of pseudobarnacles on the PDMS-UPy-Tx coatings was determined according to ASTM D5618 (2011). Five aluminous cylindrical studs (10 mm in diameter and 10 mm in height) were attached to the coating surface using epoxy adhesive (Araldite Rapid Epoxy Adhesive, Huntsman). The adhesive was cured for 3 days at ambient temperature. A constant speed force in parallel with the coatings was applied to measure the removal strength using the force gauge (SUNDOO SH-500).

Antifouling tests. The antifouling performances of PDMS-UPy-Tx were evaluated by QCM-D, antibacterial test and anti-diatom test.

(1) *Quartz crystal microbalance with dissipation (QCM-D).* The protein resistance of PDMS-UPy-Tx was investigated using a QCM-D from Q-sense AB (Sweden). A polymer film was prepared by spin-casting a PDMS-UPy-Tx solution in THF (5.0 mg mL⁻¹) onto an AT-cut quartz crystal surface and cured at room temperature for 3 days. The details of the QCM-D measurement can be found elsewhere.⁴⁰ Fibrinogen was chosen as the model protein to carry out the experiment.⁴¹ Briefly, the frequency change (Δf) is related to the protein adsorption on the quartz crystal, whereas the dissipation change (ΔD) is related to the viscoelastic properties of the adsorbed protein layer.

(2) *Antibacterial test.* Marine bacteria *Pseudomonas* sp. were incubated to evaluate the antibacterial performance of PDMS-UPy-Tx coatings. Each PDMS-UPy-Tx coating was put in a polystyrene 24-well tissue culture plate with 1 mL of bacterial

suspension and cultured at 30 °C for 4 h before the assay. Subsequently, the bacteria were stained using a LIVE/DEAD BacLight Bacterial Viability Kit, followed by gentle washing with ASW to remove weakly attached bacteria and residual dye. Adherent bacteria were observed using a fluorescence microscope (Scope A1, Zeiss).

(3) *Anti-diatom test.* *Navicula incerta* (*N. incerta*) obtained from Xiamen University was used to carry out the anti-diatom test on PDMS-UPy-Tx-coated glass slides. Guillard's F/2 culture medium was employed to cultivate *N. incerta* in a plant growth chamber (Bluepard) at 23 °C with a 12:12 h light-dark cycle. Glass slides with PDMS-UPy-Tx coatings were placed in a 24-well plate, and 1 mL of the diatom suspension (chlorophyll content: 0.50 $\mu\text{g mL}^{-1}$) was added to each well. After incubation for 24 h, the glass slides were gently rinsed with artificial seawater to remove unattached diatoms and were observed using a microscope (Scope A1, Zeiss). Ten fields of view on each of the triplicate slides were counted, and the average density value was recorded.

Results and discussion

The PDMS-UPy-Tx polymer was synthesized from the addition reaction of difunctional carbinol-terminated poly(dimethylsiloxane), amphiphilic telomer, UPy-diol, IPDI and BDO (Scheme 1 and Table 1). Incorporation of amphiphiles and UPy-diol was confirmed by the methylene group ($-\text{CH}_2-$) of the PEG moiety in 3.37 ppm in ¹H NMR and the C-N group of the UPy moiety in 1537 cm⁻¹ in FTIR spectra. The adhesion strength of coatings on substrates is crucial in different application environments. We studied the adhesion strength of PDMS-UPy-Tx on various substrates (Fig. 1). For the control sample PDMS, the adhesion strength exhibits a small value (0.3–0.4 MPa). However, PDMS-UPy-Tx coatings have comparatively higher adhesion strengths on the fiberglass epoxy resin panel ranging from 1.4 to 3.0 MPa, which can be attributed to strong hydrogen bonding with polar groups such as hydroxyl groups and ether groups of fiberglass epoxy resin.⁴² Regarding the concrete substrate, strong mechanical anchorage between the coating and the

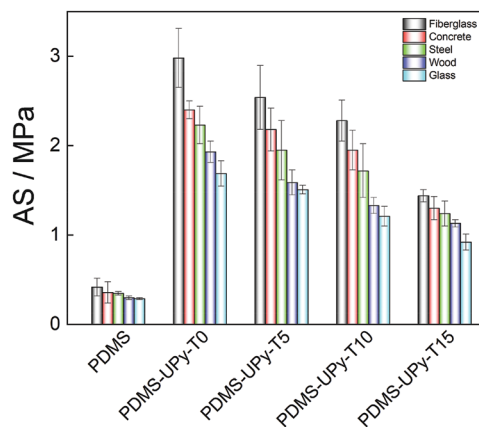


Fig. 1 Adhesion strength (AS) of PDMS-UPy-Tx on various substrates.

surface of concrete as well as hydrogen bonding leads to high adhesion strength (1.3–2.4 MPa).⁴³ Besides, the rough surface of steel and surface hydroxyl groups can interact with coatings through mechanical interlocking and hydrogen bonding,⁴⁴ contributing to a high adhesion strength on steel as well. Moreover, PDMS-UPy-Tx coatings have good adhesion strength on substrates like wood and glass (0.9–1.9 MPa). Note that the adhesion strength of PDMS-UPy-Tx on all the tested substrates decreases with increasing telomer content, and this is because the introduction of the telomer results in more side chains along the polyurethane backbones, which leads to less hydrogen bonding between coatings and substrates.⁴⁵ For the PDMS elastomer, a tie-coat is needed to improve the adhesion strength of PDMS to the substrates in practical applications,³ whereas PDMS-UPy-Tx could be applied on the substrates without a tie-coat, showing a more efficient and convenient approach. Fig. S8 (ESI†) presents the adhesion strength of PDMS-UPy-Tx on the fiberglass epoxy resin panel before and after immersion in ASW. After immersion in ASW for 24 h, the adhesion strength of PDMS-UPy-Tx decreases slightly. With increasing telomer content, a more obvious decrease was observed in the adhesion strength of PDMS-UPy-Tx. This is because high telomer content results in a more hydrophilic surface, which could lead to swelling of the coating and lowering of the toughness of the coating, resulting in a lower adhesion strength. Nevertheless, PDMS-UPy-T15 still has an

adhesion strength higher than 1 MPa, which is sufficient for applications. Overall, PDMS-UPy-Tx coatings still demonstrate good adhesion strength on various substrates compared with PDMS, since the UPy moieties can provide strong interfacial adhesion through complementary hydrogen bonding,⁴⁶ showing the potential of the PDMS-UPy-Tx coatings in various marine applications.⁴⁷

Tensile tests were carried out to examine the mechanical properties and self-healing efficiency of PDMS-UPy-Tx (Fig. 2a, Table 2 and Table S1, ESI†). PDMS-UPy-T0 shows the highest tensile strength (5.34 MPa) and toughness (17.10 MJ m⁻³) but the lowest self-healing efficiency at 25 °C. This is because the incorporation of UPy endows the polymer with high mechanical strength while restricts the mobility of polymer chains to some extent.⁴⁸ Increasing telomer content lowers the tensile strength (4.43 MPa for PDMS-UPy-T5, 1.91 MPa for PDMS-UPy-T10 and 0.31 MPa for PDMS-UPy-T15) and toughness (15.36 MJ m⁻³ for PDMS-UPy-T5, 11.18 MJ m⁻³ for PDMS-UPy-T10 and 2.86 MJ m⁻³ for PDMS-UPy-T15) but improves the elasticity of PDMS-UPy-Tx, where PDMS-UPy-T15 can be stretched to more than 1000% of its original length. Moreover, as the content of the telomer increases, the self-healing efficiency increases and PDMS-UPy-T15 shows full recovery of mechanical strength of the polymer, indicating the highest self-healing efficiency (~100%). This is probably because the hydrogen bonding interaction of polymer chains is disrupted

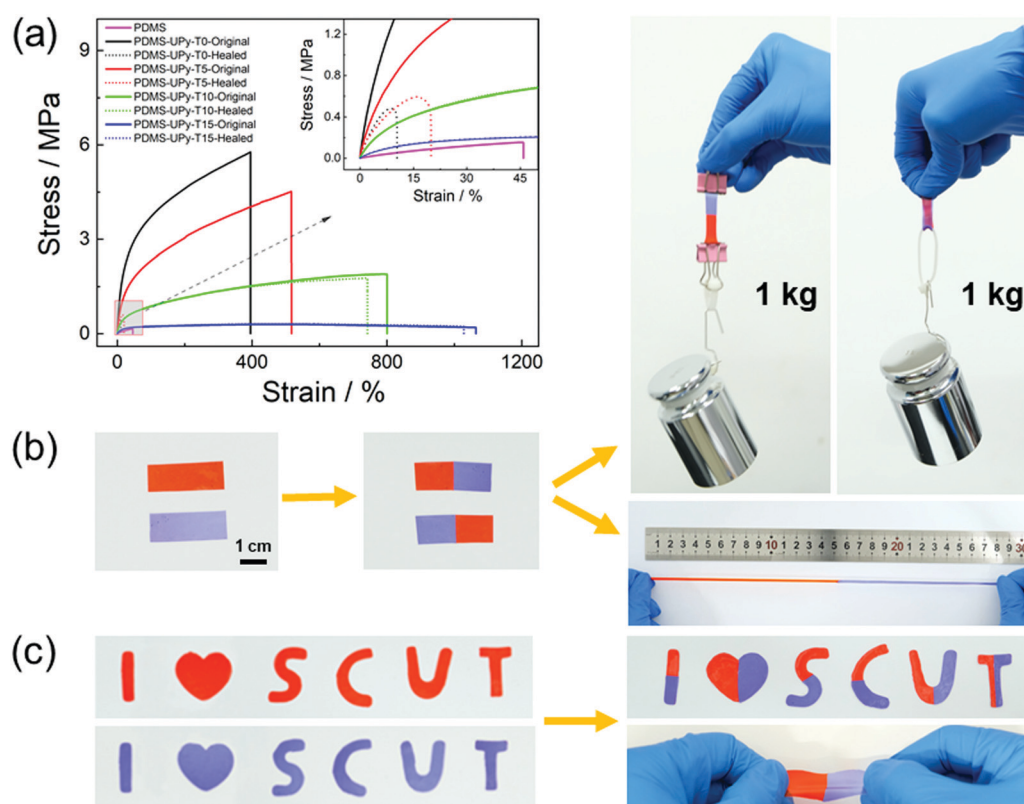


Fig. 2 (a) Stress–strain curves of original and healed PDMS-UPy-Tx (the inset figure details the former part of the curves). (b) Healed PDMS-UPy-T15 polymer films can be stretched to 1000% of their original length and withstand 1 kg weight from the stretching or bending direction. (c) Images of different combined shapes of healed PDMS-UPy-T15 samples. The healing took place at room temperature for 48 h.

Table 2 Mechanical properties of PDMS-UPy-Tx

Sample	Tensile strength (MPa)		Elongation (%)		Elastic modulus (MPa)	
	Original	Healed ^a	Original	Healed ^a	Original	Healed ^a
PDMS	0.16 ± 0.01	—	47 ± 7	—	0.57 ± 0.04	—
PDMS-UPy-T0	5.34 ± 0.13	0.40 ± 0.08	394 ± 15	10 ± 3	15.77 ± 0.43	8.80 ± 0.53
PDMS-UPy-T5	4.43 ± 0.12	0.66 ± 0.07	529 ± 56	18 ± 3	9.63 ± 0.14	7.36 ± 0.80
PDMS-UPy-T10	1.91 ± 0.09	1.80 ± 0.05	800 ± 36	747 ± 6	4.37 ± 0.34	4.27 ± 0.05
PDMS-UPy-T15	0.31 ± 0.02	0.31 ± 0.01	1078 ± 49	1029 ± 3	1.44 ± 0.04	1.41 ± 0.02

^a The sample was cut, reconnected and allowed to heal at room temperature for 48 h.

by the F-PEG side chains and incorporation of the telomer lowers the glass temperature transition (T_g) of the polymer (Table 1). Actually, the self-healing efficiency can be improved by increasing the temperature. Fig. S9 (ESI[†]) shows the self-healing ability of PDMS-UPy-T0. Clearly, the healing efficiency of PDMS-UPy-T0 improves with increasing temperature (Table S2, ESI[†]). When the temperature is raised up to 80 °C, PDMS-UPy-T0 almost restores the original mechanical strength. This is because elevated temperature accelerates the dissociation of quadruple hydrogen bonds, which brings about enhanced chain mobility and heals the crack. Besides, the re-association of quadruple hydrogen bonds between UPy dimers recovers the mechanical strength of the healed polymer.⁴⁹ Overall, the UPy moieties play a critical role in crack healing and mechanical property restoration of PDMS-UPy-Tx.

Although the introduction of the telomer compromises the mechanical strength of the polymer, PDMS-UPy-T15 has higher tensile strength, room-temperature self-healing ability and elasticity compared with unrepairable PDMS. The self-healing performance of PDMS-UPy-T15 was further examined using macroscopic healing tests (Fig. 2b and c). The rectangular PDMS-UPy-T15 (0.3 g) polymer films show high elasticity and sufficient strength, and they can be stretched to 1000% of their original length and withstand 1 kg weight from the stretching or bending direction (Fig. 2b). Also, separated parts of the polymer with different shapes can be recombined and healed into a new combination, which can keep integrity upon an external force (Fig. 2c). In a word, with quadruple hydrogen bonds of UPy moieties, the polymer exhibits good self-healing performance at room temperature with excellent elasticity and sufficient mechanical strength.

Cyclic tensile tests were also conducted to evaluate the dissipation energy of PDMS-UPy-Tx (Fig. S10, ESI[†]). According to the first hysteresis loops of cyclic tensile tests, the energy dissipation of silicon-based polyurethane is 5.69 MJ m⁻³ for PDMS-UPy-T0, 3.81 MJ m⁻³ for PDMS-UPy-T5, 1.17 MJ m⁻³ for PDMS-UPy-T10 and 0.20 MJ m⁻³ for PDMS-UPy-T15. Clearly, for each sample, a significant hysteresis was observed in the first loading-unloading cycle, indicative of great energy dissipation through the rupture of hydrogen bonds.⁵⁰ Besides, there is a significant decrease in the amount of energy dissipation in cycle 2 compared to cycle 1, whereas the energy dissipation decreases slightly from cycle 3 to cycle 10. This is probably because a part of the ruptured hydrogen bonds cannot reform within the time scale of each cycle, so fewer hydrogen bonds

continue to dissipate energy in the following successive cycle.⁵¹ Overall, the strong hydrogen bonding interaction of PDMS-UPy-Tx provides toughness and sufficient energy dissipation when the materials are deformed by an external force, which also contribute to improved adhesion strength.⁵²

Fig. 3 shows the self-healing process of damaged PDMS-UPy-Tx coatings at room temperature in ASW. For PDMS-UPy-T0, the crack was clearly visible after 48 h, indicating that the coating cannot heal. The scratch in PDMS-UPy-T5 slightly healed with time, but could not heal completely. The healing ability increased as the telomer content increased, with PDMS-UPy-T10 and PDMS-UPy-T15 completely healed after 24 and 12 h, respectively. The self-healing property can be attributed to the chain mobility and quadruple hydrogen bonds formed among the UPy moieties in the area with repaired damage. Note that the healing efficiency is affected by the telomer content. As mentioned above, higher telomer content in PDMS-UPy-T10 and PDMS-UPy-T15 results in lower hard segments in polyurethane so that higher chain mobility can be achieved.⁵³ Moreover, a higher telomer content leads to increased steric hindrance. This is because increasing telomer content results in more side chains of PEG and fluorocarbon segments along the main chains of polyurethane, which also reduce the hydrogen bonding interaction between polymer chains.⁵⁴ Both factors contribute to the decreased restriction of the polymer chains and the increased contact of the UPy moieties, resulting in the formation of sufficient quadruple hydrogen bonds in the area with repaired damage to give a higher healing efficiency. We also examined the self-healing ability of coatings in air (Fig. S11, ESI[†]). The coatings with 10 and 15 wt% telomer again showed excellent self-healing abilities. This means that the rapid self-healing process can occur in water or air, making the polymer useful for marine antifouling or biomedical applications.⁵⁵

The surface wettability of PDMS-UPy-Tx coatings was examined by water contact angle measurements (Fig. 4a and b). Before immersion in ASW, PDMS-UPy-Tx coatings have initially high WCA (105°–110°) and low SE (20–22 mJ m⁻²) due to the hydrophobic nature of PDMS. Increasing telomer content increases WCA and decreases SE, indicative of the enrichment of the fluorocarbon-containing telomer on the coating surface. After immersion, a significant decrease in WCA and an increase in SE were observed in coatings with increased telomer content, which is attributed to the exposure of hydrophilic PEG chains on the surface of coatings during the

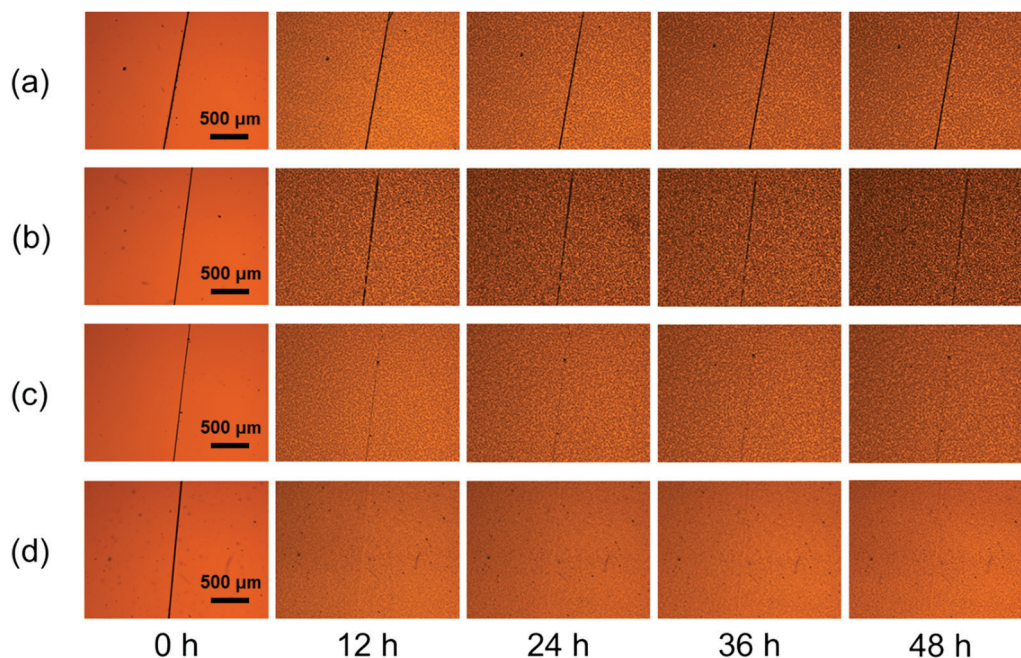


Fig. 3 Self-healing process of (a) PDMS-UPy-T0, (b) PDMS-UPy-T5, (c) PDMS-UPy-T10, and (d) PDMS-UPy-T15 at room temperature in ASW.

water-driven reconstruction of the amphiphilic telomer.⁵⁶ The dynamic WCA of these coatings was also examined (Fig. S12, ESI†). Before immersion, the coatings have an increasing advancing WCA and a decreasing receding WCA with increasing telomer content. This is probably because the hydrophobic

fluorocarbon segments increase the advancing WCA while the hydrophilic PEG segments lower the receding WCA.⁵⁷ Likewise, surface rearrangement of the telomer leads to a decrease in both advancing WCA and receding WCA upon immersion. Even though the incorporation of the amphiphilic telomer changes

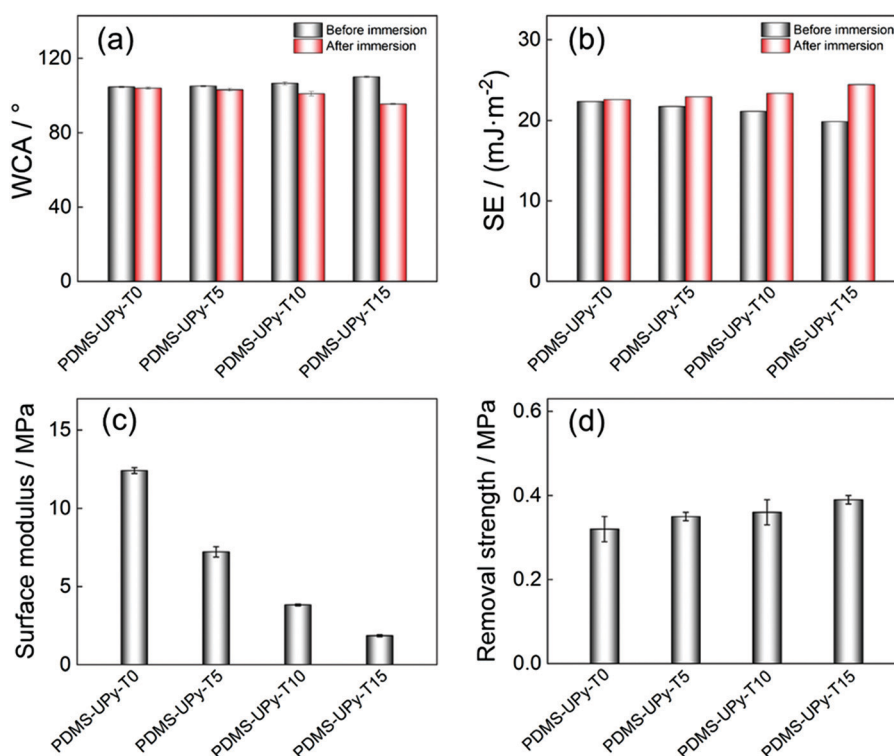


Fig. 4 (a) Water contact angle (WCA) and (b) surface energy (SE) of PDMS-UPy-Tx coatings before and after immersion in ASW. (c) Surface modulus of PDMS-UPy-Tx coatings. (d) Removal strength of pseudobarnacles adhered on PDMS-UPy-Tx coatings.

the wettability of the coating surface, the coatings still have low SE after immersion in ASW, which is beneficial for fouling release.

Fig. 4c shows the surface modulus of the PDMS-UPy-Tx coatings measured by AFM (force-depth curves shown in Fig. S13, ESI†). PDMS-UPy-T0 has the highest surface modulus of 12.4 MPa, demonstrating that incorporation of UPy and urethane leads to a high surface modulus. As the telomer content increases, the surface modulus decreases. This is because higher telomer content leads to lower hard segments in polyurethane and thereby decreases the surface modulus. For PDMS-UPy-T15, it has a surface modulus of ~ 1.9 MPa, close to the reported data of PDMS,⁵⁸ indicating that the coating with such a low modulus could have good fouling release ability.

Generally, the fouling release ability of coatings correlates with the elastic modulus and surface energy.³ The fouling release performance of PDMS-UPy-Tx coatings was evaluated by the removal strength of pseudobarnacles (Fig. 4d). PDMS-UPy-T0 has a removal strength of 0.32 MPa. Actually, an elastic surface can facilitate the failure of adherence because more micro scratches occur upon an external force.⁵⁹ When the telomer content increases to 10 wt%, the surface becomes somewhat tacky rather than elastic, which does not favor the break of the adhesive interface.⁶⁰ Therefore, the removal strength slightly increases with the telomer content. Nevertheless, these coatings still have good fouling release performance, which is similar to that of other silicone-based coatings.⁶¹

The antifouling performances of PDMS-UPy-Tx were evaluated by a protein resistance test, an antibacterial test and an anti-diatom test. The protein resistance of as-prepared coatings was determined by a QCM-D test. With a high molecular weight ($M_w = 340$ kDa), Fibrinogen has a higher propensity to adsorb to different types of surfaces and adheres more strongly than most other proteins. Therefore, Fibrinogen has been widely chosen as the model protein in the protein-resistance test of marine antifouling materials.^{5,41,62} Fig. 5a shows the time dependence of the frequency (Δf) and dissipation shifts (ΔD) of Fibrinogen on PDMS-UPy-Tx coatings. PDMS-UPy-T0 shows a significant decrease in Δf and an increase in ΔD after rinsing with ASW, indicating the strong adsorption of Fibrinogen. PDMS-UPy-T5 also shows an obvious decrease in Δf and an increase in ΔD , indicating limited protein resistance. By contrast, PDMS-UPy-T10 demonstrates better protein resistance, reflected in the lower decrease in Δf and the increase in ΔD . For PDMS-UPy-T15, changes in Δf and ΔD are almost negligible after rinsing, showing exceptional protein resistance and indicating that protein resistance improves with increasing telomer content.

Fig. 5b shows the antibacterial performance of the PDMS-UPy-Tx coatings after exposure to the marine bacteria *Pseudomonas* sp. A large number of bacteria adhered to a silicon wafer control with no antibacterial components. For PDMS-UPy-T0 samples, a significant number of bacteria adhered to the coating surface (86% relative bacterial adhesion (RBA)), indicating that they are highly subjected to bacterial adhesion. As the telomer content increases, the number of adherent bacteria

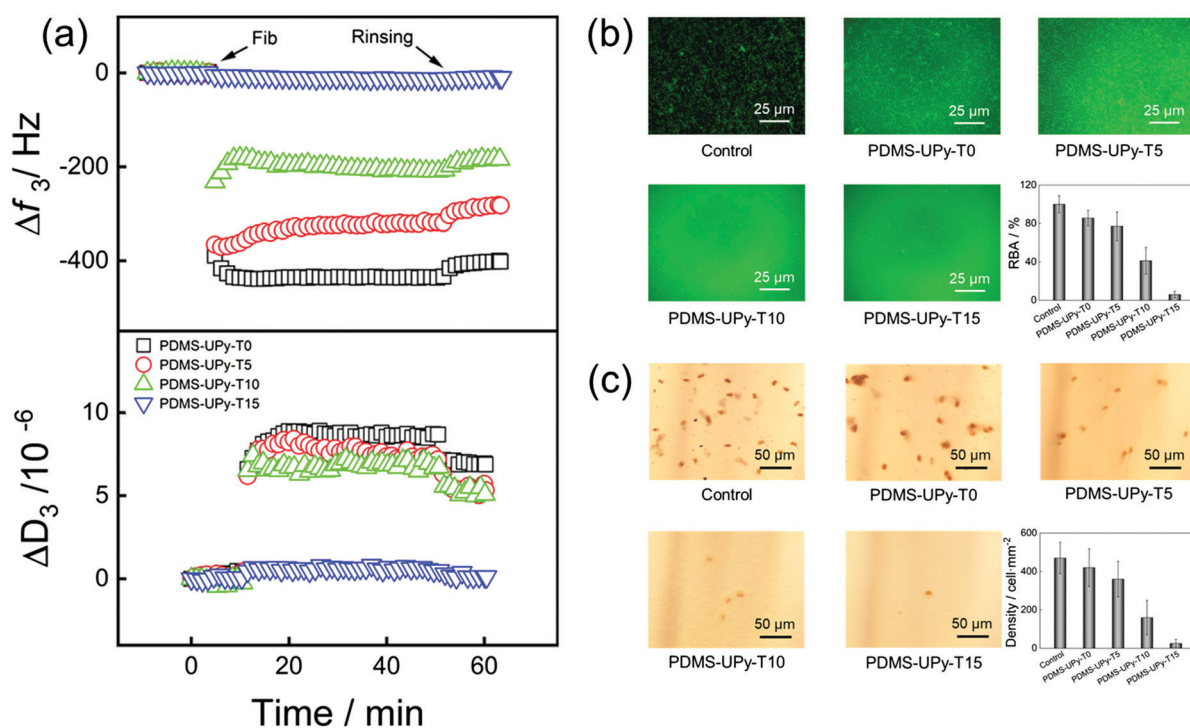


Fig. 5 Antifouling performances of PDMS-UPy-Tx coatings. (a) Time dependence of the frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of Fibrinogen. (b) Fluorescence images of adherent *Pseudomonas* sp. and relative bacterial adhesion (RBA). (c) Optical images of adherent diatom *N. incerta* and the qualified diatom density.

clearly decreases. PDMS-UPy-T15 has an RBA of only 6% because the PEG groups from the surface-enriched telomer improve bacterial resistance. These results show that the coatings with a high telomer content exhibit excellent antibacterial performance.

Fig. 5c shows the optical images of adherent *N. incerta* on PDMS-UPy-Tx coatings and their density. The silicon wafer control demonstrates the high fouling pressure environment. A large number of diatoms were observed on PDMS-UPy-T0 (440 cell m^{-2}) as the diatoms prefer hydrophobic surfaces.⁶³ The density of diatoms remarkably decreased from 360 cell m^{-2} for PDMS-UPy-T5 to 25 cell m^{-2} for PDMS-UPy-T15 as the telomer content increased, indicating that PDMS-UPy-Tx coatings have advanced anti-diatom performance with high telomer content on the surface of coatings. These results are consistent with those of the QCM-D and antibacterial tests, further demonstrating the high fouling resistance of the PDMS-UPy-T15 coating.

Conclusions

We have prepared a novel self-healing and highly elastic silicone-based polyurethane with UPy and amphiphilic pendant chains. Introduction of the UPy moieties formed quadruple hydrogen bonding, which provided excellent self-healing ability and high elasticity to the polymer. Furthermore, the UPy moieties and urethane endow the coatings with improved adhesion on various substrates. The chemically tethered amphiphiles can effectively resist the adsorption of protein, marine bacteria, and diatoms. In comparison to silicone coatings with physically mixed antifoulants, the coatings are more environmentally friendly and potentially more durable. In brief, silicone-based antifouling coatings with improved adhesion, self-healing ability, high elasticity and antifouling performance were successfully developed, and they are promising as novel anti-biofouling coatings.

Conflicts of interest

There are no conflicts to declare.

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